

Hypothesis Testing for Population Mean

Statistics Worksheet · Grade 11–12 / Introductory College Statistics

Name: _____

Date: _____

Learning Objectives

- State null and alternative hypotheses for a population mean using correct notation
- Calculate a z-test statistic and find the corresponding P-value
- Make a conclusion about the null hypothesis by comparing the P-value to a given significance level

Problems

1. A teacher claims the mean score on a standardized math test is 75. Write the null hypothesis and the alternative hypothesis if a researcher believes the true mean is different from 75. Use μ to represent the population mean.

$$H_0 : \mu = 75 \quad \text{vs.} \quad H_a : \mu \neq 75$$

2. A nutritionist believes that adults consume MORE than 2000 calories per day on average. Write the null and alternative hypotheses for this claim using μ .

$$H_0 : \mu = 2000 \quad \text{vs.} \quad H_a : \mu > 2000$$

3. A researcher wants to test whether the mean resting heart rate for athletes is less than 65 beats per minute. State both hypotheses and identify the type of test (left-tailed, right-tailed, or two-tailed).

$$H_0 : \mu = 65 \quad \text{vs.} \quad H_a : \mu < 65$$

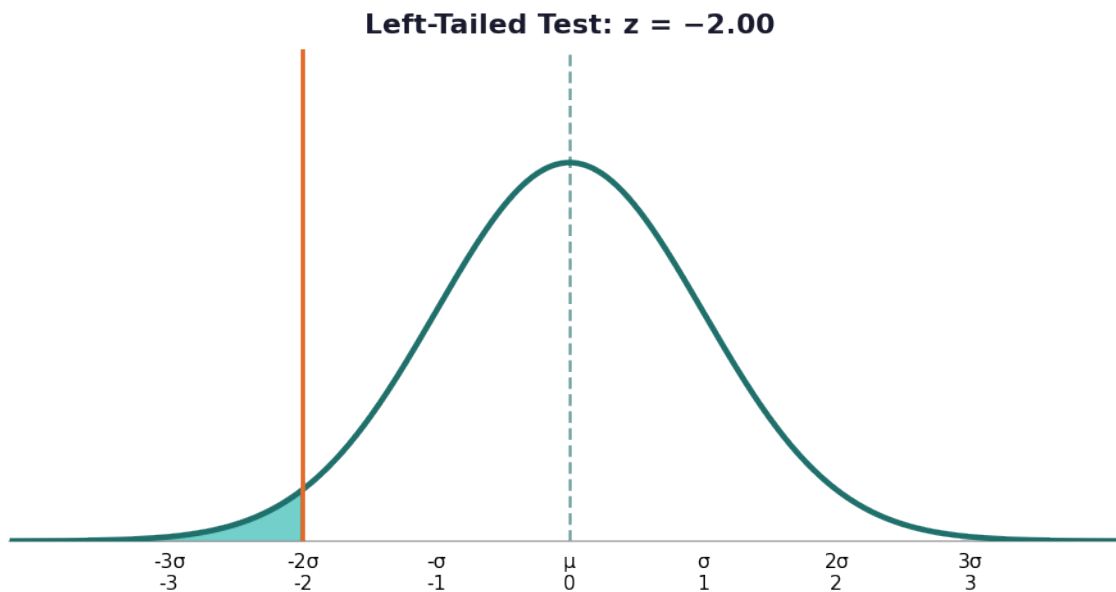
4. Before conducting a hypothesis test for a population mean, you must check three conditions. A researcher randomly selects 40 students from a large university, and the population of scores is stated to be normally distributed. The sample size is less than 10% of the total student population. State whether all three conditions are satisfied and identify each condition.

5. The population mean weight of a cereal box is claimed to be 400 grams with a known population standard deviation of 10 grams. A quality inspector randomly selects 25 boxes and finds a sample mean of 396 grams. Calculate the z-test statistic.



$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

6. Using the z-test statistic of negative 2.00 from the cereal box problem above, find the P-value for a left-tailed test. The shaded area below shows the region of interest. Then state whether you reject or fail to reject the null hypothesis at a significance level of 0.05.

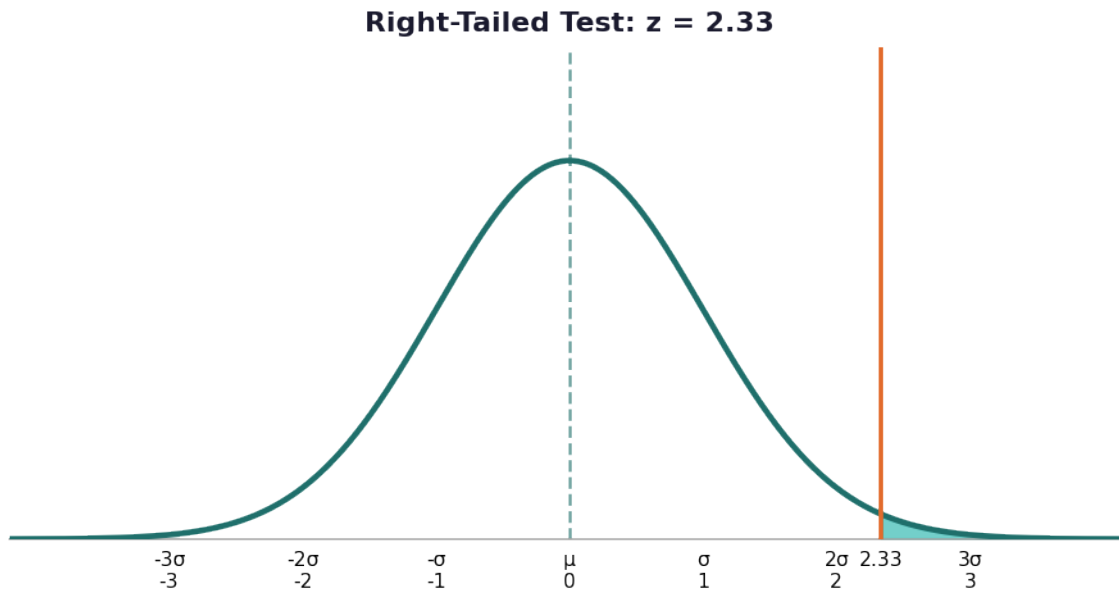


7. The mean score for US college students on the Survey of Study Habits and Attitudes (SSHA) is 115 with a known population standard deviation of 30. Monica suspects that older students (age 30 and above) have better attitudes toward school. She randomly selects 25 such students and finds a sample mean score of 118.6. At a significance level of 0.05, carry out all five steps of the hypothesis test.

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

8. A hospital claims the average patient wait time is 30 minutes. A health inspector believes the actual wait time is longer. She samples 36 patients and finds a mean wait time of 33.5 minutes. The population standard deviation is known to be 9 minutes. Conduct a full hypothesis test at alpha equals 0.01.



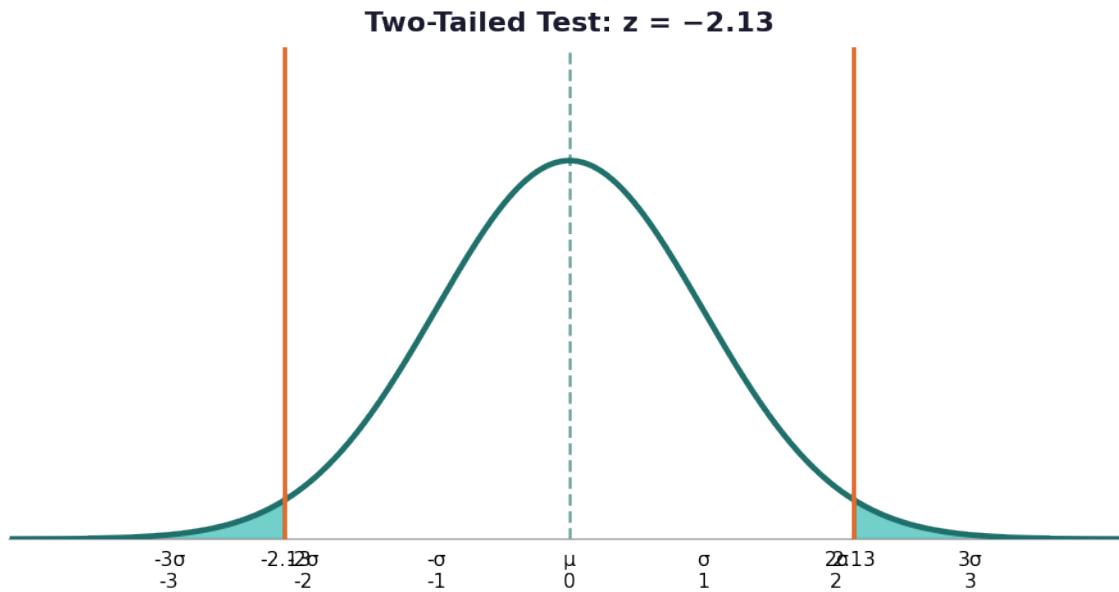


9. The table below summarizes three hypothesis tests. For each test, determine whether to reject or fail to reject the null hypothesis based on the given P-value and significance level. Fill in the Decision and Conclusion columns.

Test	P-value	Alpha (α)	Decision	Conclusion
A	0.032	0.05		
B	0.078	0.05		
C	0.004	0.01		

10. A manufacturer claims that the mean lifetime of their light bulbs is 1500 hours with a known population standard deviation of 120 hours. A consumer group believes the mean lifetime is actually different from 1500 hours. They randomly test 64 bulbs and find a sample mean lifetime of 1468 hours. The population is normally distributed. Perform a complete hypothesis test at alpha equals 0.05, compute the z-statistic and P-value, and write a full conclusion in the context of the problem.





Hypothesis Testing for Population Mean — Answer Key

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Answer Key

1. Answer: $H_0: \mu = 75$; $H_a: \mu \neq 75$ (two-tailed test)

- The null hypothesis always states equality: $H_0: \mu = 75$.
- Because the researcher believes the mean is DIFFERENT (not specifically higher or lower), the alternative is two-tailed: $H_a: \mu \neq 75$.

2. Answer: $H_0: \mu = 2000$; $H_a: \mu > 2000$ (right-tailed test)

- The null hypothesis is: $H_0: \mu = 2000$ (no change from the assumed value).
- Because the claim is that adults consume MORE than 2000, the alternative is right-tailed: $H_a: \mu > 2000$.

3. Answer: $H_0: \mu = 65$; $H_a: \mu < 65$ — Left-tailed test

- The null hypothesis states the mean equals the claimed value: $H_0: \mu = 65$.
- Because the researcher suspects the mean is LESS than 65, the alternative is left-tailed: $H_a: \mu < 65$.
- This is a left-tailed test because the rejection region is in the left tail.

4. Answer: All three conditions are satisfied: (1) Random sample ✓, (2) Normal population ✓, (3) Independence ($n < 10\%$ of population) ✓

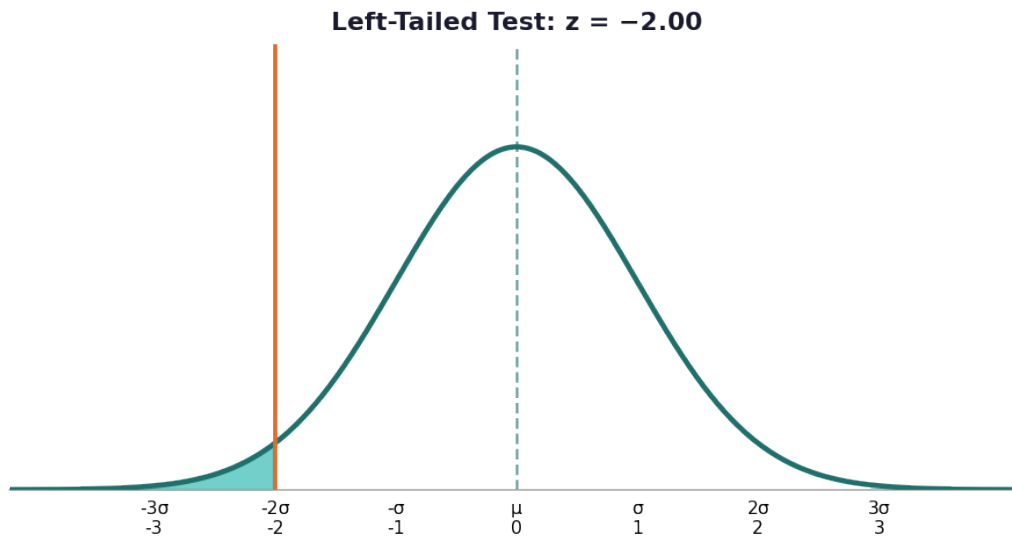
- Condition 1 — Random: The problem states the sample was randomly selected. ✓
- Condition 2 — Normality: The population is stated to be normally distributed. ✓
- Condition 3 — Independence: The sample of 40 is less than 10% of the large university population. ✓
- Since all three conditions are met, we may proceed with hypothesis testing without caution.

5. Answer: $z = -2.00$

- Identify values: $x = 396$, $\mu = 400$, $\sigma = 10$, $n = 25$.
- Calculate standard error: $\sigma / \sqrt{n} = 10 / \sqrt{25} = 10 / 5 = 2$.
- Calculate z : $z = (396 - 400) / 2 = -4 / 2 = -2.00$.
- The z -test statistic is -2.00 , indicating the sample mean is 2 standard errors below the claimed mean.

6. Answer: P-value ≈ 0.0228 ; Since $0.0228 < 0.05$, reject H_0 .





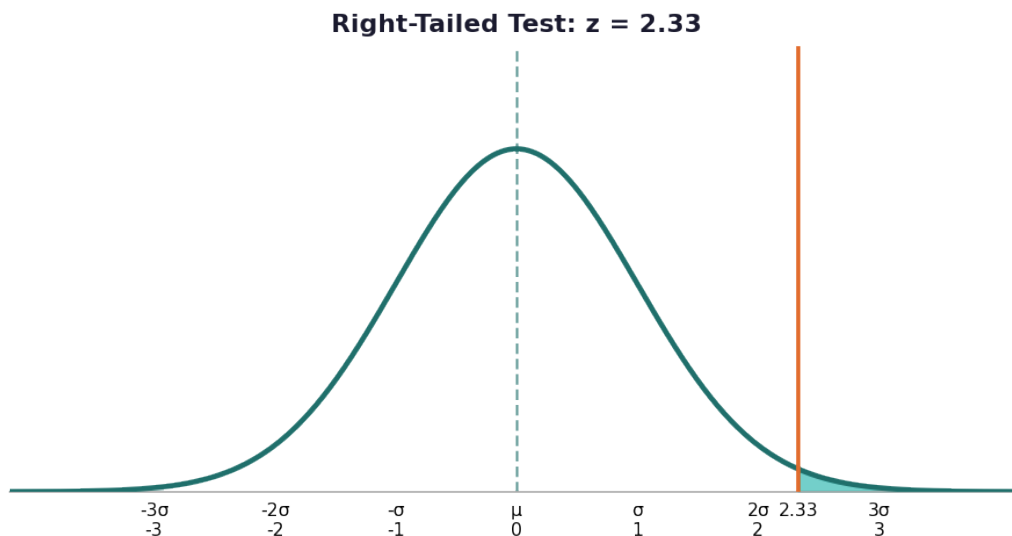
- For a left-tailed test with $z = -2.00$, find $P(Z < -2.00)$ using the standard normal table.
- $P(Z < -2.00) \approx 0.0228$.
- Compare: $P\text{-value} = 0.0228 < \alpha = 0.05$.
- Decision: Since the P-value is less than α , reject the null hypothesis.
- Conclusion: There is sufficient evidence that the mean cereal box weight is less than 400 grams.

7. Answer: $z \approx 0.60$; P-value ≈ 0.2743 ; Fail to reject H_0 . Insufficient evidence that older students score higher.

- Step 1 — Hypotheses: $H_0: \mu = 115$; $H_a: \mu > 115$ (right-tailed, Monica suspects higher scores).
- Step 2 — Conditions: Random sample ✓; Population normally distributed ✓; $n = 25 < 10\%$ of all students ✓.
- Step 3 — Test statistic: $z = (118.6 - 115) / (30 / \sqrt{25}) = 3.6 / 6 = 0.60$.
- Step 4 — P-value: $P(Z > 0.60) = 1 - 0.7257 = 0.2743$.
- Step 5 — Conclusion: $0.2743 > 0.05$, so we fail to reject H_0 . There is not sufficient evidence that older students have a higher mean SSHA score than the general population.

8. Answer: $z \approx 2.33$; P-value ≈ 0.0099 ; Reject H_0 at $\alpha = 0.01$. Evidence supports longer wait times.





- Step 1 — Hypotheses: $H_0: \mu = 30$; $H_a: \mu > 30$ (right-tailed).
- Step 2 — Conditions: Random sample ✓; Assume population is normally distributed ✓; $n = 36 < 10\%$ of all patients ✓.
- Step 3 — Test statistic: $z = (33.5 - 30) / (9 / \sqrt{36}) = 3.5 / 1.5 \approx 2.33$.
- Step 4 — P-value: $P(Z > 2.33) = 1 - 0.9901 = 0.0099$.
- Step 5 — Conclusion: $0.0099 < 0.01$, so we reject H_0 . There is sufficient evidence at the 0.01 level that the mean patient wait time exceeds 30 minutes.

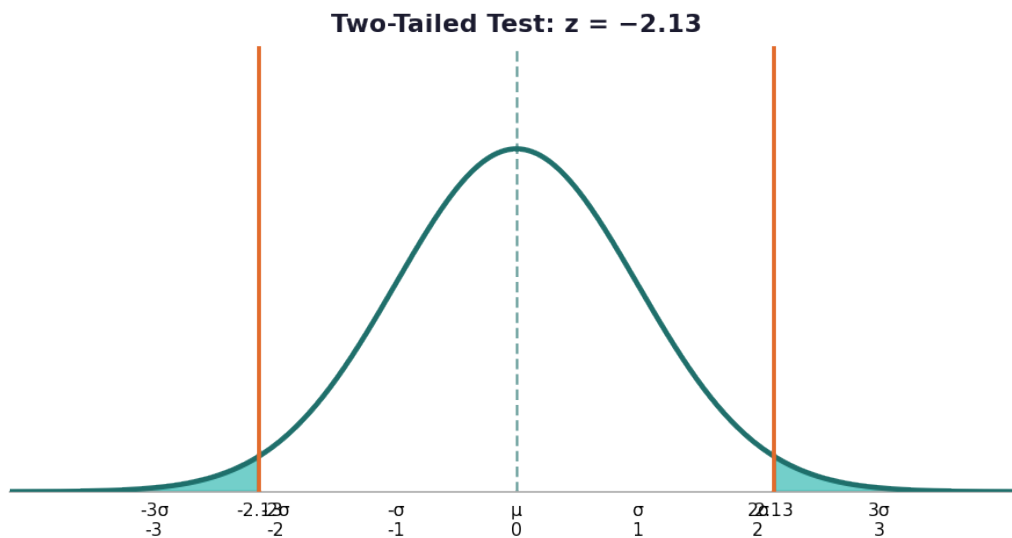
9. Answer: Test A: Reject H_0 ; Test B: Fail to reject H_0 ; Test C: Reject H_0

Test	P-value	Alpha (α)	Decision	Conclusion
A	0.032	0.05	Reject H_0	Sufficient evidence against H_0
B	0.078	0.05	Fail to reject H_0	Insufficient evidence against H_0
C	0.004	0.01	Reject H_0	Sufficient evidence against H_0

- Rule: If P-value $\leq \alpha$, reject H_0 . If P-value $> \alpha$, fail to reject H_0 .
- Test A: $0.032 \leq 0.05 \rightarrow$ Reject H_0 .
- Test B: $0.078 > 0.05 \rightarrow$ Fail to reject H_0 .
- Test C: $0.004 \leq 0.01 \rightarrow$ Reject H_0 .

10. Answer: $z \approx -2.13$; P-value ≈ 0.0332 ; Reject H_0 . Evidence that mean bulb lifetime differs from 1500 hours.





- Step 1 — Hypotheses: $H_0: \mu = 1500$; $H_a: \mu \neq 1500$ (two-tailed, consumer group believes it's different).
- Step 2 — Conditions: Random sample ✓; Population normally distributed ✓; $n = 64 < 10\%$ of all bulbs produced ✓. All conditions met.
- Step 3 — Test statistic: $SE = 120 / \sqrt{64} = 120 / 8 = 15$. $z = (1468 - 1500) / 15 = -32 / 15 \approx -2.13$.
- Step 4 — P-value (two-tailed): $P = 2 \times P(Z < -2.13) = 2 \times 0.0166 = 0.0332$.
- Step 5 — Conclusion: Since P-value = 0.0332 < $\alpha = 0.05$, we reject H_0 . At the 5% significance level, there is sufficient evidence to conclude that the mean lifetime of the manufacturer's light bulbs is significantly different from 1500 hours.

